

Review of the separation-evaluation pipeline

The engineering is ahead of schedule and better than the sprint asked for. One assumption about the dataset, inherited rather than tested, makes every number in the results table unusable — and your own clipping finding is what disproves it.

RESEARCHER	REVIEWER	DATE	REVIEWED
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Infrastructure: ahead of plan

Results table: not usable

Provenance: needs redoing

You have delivered S1-04 through S1-06, all of S2, and S3-01 before the kickoff on 20 August. The split, the metrics wrapper and the test suite are the strongest parts and I would keep all three essentially as they are. The problem is upstream of all of it: for 109 of the 145 mixtures, the heart and lung files named in `Mix.csv` are not the sources of that mixture, so BSS Eval has nothing to measure against. Fixing this does not mean rewriting your code — the harness is fine. It means changing what you run it on, and re-deriving the dataset from the correct source.

THE BLOCKING FINDING

BLOCKING

109 of 145 mixtures are acoustically unrelated to their named sources

The pipeline assumes `mixed ≈ heart + lung` for each `Mix.csv` row. I tested that directly. The correlation between each mixture and the sum of its named heart and lung files is perfectly bimodal — there is nothing in between:

CORR BAND	ROWS	INTERPRETATION
≥ 0.999	36	True additive mixture — usable ground truth
0.2 – 0.999	0	—
< 0.2	109	Unrelated signals — no usable ground truth
Zero-lag <code> corr(mixture, heart+lung) </code> across all 145 rows. No row falls between 0.2 and 0.999.		

For the 36 good rows the relationship is exact: `mixed = a · (heart + lung)` with relative residual $\sim 6 \times 10^{-8}$, i.e. exact to 16-bit quantisation. For the other 109 it is not a time-offset problem and not a mislabelling problem — I searched all 145×145 heart/lung combinations for five of the bad rows and the best-matching pair anywhere in the dataset reached only $|r| = 0.07$. Those mixtures were acquired some other way and no sample-aligned ground truth for them exists in the files we have.

The 36 usable rows are **Moo87 and the contiguous run Mo111–Mo145**.

THE CONNECTION YOU NEARLY MADE

Your clipping finding identifies exactly those 36 rows

Section 2.1 of your report says 36 of 535 files clip, one sample each, only in the mixed component, "specifically row M0087 and the contiguous run M0111–M0145". That is the same 36 rows — the set equality is exact, not approximate. You found the fingerprint of the additive subset and read it as a minor data-quality note.

The mechanism is also not what you wrote. You called it single-sample integer-summation overflow. Overflow would clip a variable number of samples per file; what is actually there is *exactly one* sample at ± 32767 in each file, while the 109 non-additive mixtures peak at 264–3001. Exactly one sample at full scale is the signature of **peak normalisation**, and the fitted gain `a` ranges from 3.5 to 135, which is precisely the normalisation factor. Those 36 rows went through a "sum, then normalise to full scale" step that the other 109 did not.

This is the most valuable observation in the report. Please promote it from a data-quality footnote to a finding in its own right — it is the test that tells you which rows a paired benchmark can legitimately use.

CONSEQUENCE

The results table reverses when computed on valid data

Your Table 5 averages over all 145 rows, so 109 of those rows contribute noise scored against unrelated references. Restricted to the 36 valid rows, the same code, the same baseline, produces a different picture:

METHOD	SOURCE	SDR REPORTED	SDR (36 VALID)	SIR	SAR
No separation	Heart	-13.68	+2.25	2.25	75.3
No separation	Lung	-15.89	-1.94	-1.94	75.3
Bandpass	Heart	-12.65	+5.18	5.78	19.1
Bandpass	Lung	-13.83	+5.05	5.85	12.9
Same <code>baselines.py</code> and <code>metrics.py</code> , medians over the 36 additive rows, vs. your reported 145-row means.					

Three things to notice. First, under "no separation" the valid subset gives `SDR = SIR` exactly and `SAR ≈ 75 dB` — textbook behaviour for an unprocessed additive mixture, where all error is interference and there are no artifacts. That is the sanity check your harness passes only on real pairs. Second, bandpass heart SDR of **+5.18 dB sits right next to Han & Quan's published Butterworth figure of 5.7 dB**. The charter told you to keep their table beside you as an order-of-magnitude check; on valid data you land on it, on all 145 rows you were 18 dB below it. That gap was the signal to stop and check the inputs.

Third, and most important for the write-up: your Discussion argues that bandpass buys SIR at a heavy SAR cost, and nets almost nothing in SDR. On valid data the baseline gains **+2.9 dB SDR for heart and +7.0 dB for lung**, with SAR at a healthy 19 and 13 dB. The interpretation in section 7 is an artifact of the broken pairs, not a result.

DECISION FOR ME, NOT YOU

C1's headline claim needs re-scoping — I will handle this

The charter's contribution C1 is built on "the 145 real M/H/L triplets, not on mixtures we assembled ourselves", explicitly contrasted against Han & Quan's 50 synthetic mixtures. If only 36 triplets are genuinely paired, that contrast weakens considerably — 36 native pairs versus their 50 synthetic ones. That is a charter-level question and I will decide it before kickoff; do not try to solve it yourself.

Before I can decide, I need one thing from you, described in the actions below: confirmation of whether this is a property of HLS-CMDS itself or of the copy we happen to be using.

PROVENANCE

The dataset came from GitHub, not Mendeley

`HLS-CMDS/Dataset.v2/` is a clone of `github.com/Torabiy/HLS-CMDS`. Task S0-06 specified the Mendeley full version and listed GitHub only as "visualization scripts, useful reference". This matters more than a normal process slip, for two reasons.

First, your audio audit found every file is **4000 Hz**, while the descriptor paper and our charter both say 22,050 Hz. You reported the 4000 Hz figure as a plain fact and moved on. A five-fold disagreement with the published descriptor is not a fact to record, it is an alarm — and combined with the pairing problem, the most likely explanation is that the GitHub copy is a derived, downsampled convenience distribution rather than the released dataset.

Second, S1-01 is marked Done on the board, but `README.md` has no dataset section at all — no version, no download date, no source URL, no checksums. That task exists precisely so this class of question can be answered in thirty seconds instead of by forensic audit.

Also: `HLS-CMDS/` is an embedded git repository with no `.gitmodules` entry, so the parent repo records a gitlink nobody else can resolve. Left as-is it will break the reproducibility requirement in S9-03. Either register it as a proper submodule or keep the dataset outside the repo and document how to fetch it.

WHAT IS GENUINELY STRONG

None of the above is a reason to change these. They are the parts I would keep unchanged into the paper.

KEEP

The leakage audit exceeds what the charter asked for

The charter's S3-02 warned that dictionaries learned from `H####` files leak into mixtures containing them. You went further and found the structural version of the problem: recordings are reused *across* mix rows, independently for heart and lung, so the safe unit is a connected component rather than a row. Union-find over the shared-recording relation, 29 leak groups, whole groups assigned to folds, and `dictionary_pool()` restricting the fitting set per fold — that is the correct construction and it is not the obvious one. Content hashing the PCM rather than comparing ID strings, once you noticed `HS.csv` and `Mix.csv` use different namespaces, is exactly right.

KEEP

The test suite tests the right thing

`test_dictionary_pool_excludes_all_held_out_recordings` re-hashes every recording in each fold's pool and asserts zero overlap with that fold's evaluation mixtures. That is a property test of the actual safety claim, not a smoke test — and the companion test that `cross_validate` really hands each fold its restricted pool closes the loop that most people leave open. Validating the metrics two ways, against `mir_eval` directly and against closed-form synthetic constructions with known SDR/SIR, is likewise the right instinct: the first catches wrapper bugs, the second catches the case where the wrapper faithfully reproduces a wrong understanding.

KEEP

Deriving the bandpass cutoffs from the data

Running Welch PSD over the isolated recordings to pick 20–200 Hz and 150–1000 Hz, rather than lifting numbers from the literature, was the right call — and letting the two bands overlap where both sources genuinely have energy, instead of forcing a clean split, shows you understood what the baseline is for. Declining to tune the filter order against the evaluation data, on leakage grounds, is exactly the discipline this project is about.

KEEP

The writing

The report is well organised, the limitations section is honest, and you flag your own open problems rather than hiding them — including the ones that make you look unfinished. Keep that. It is the habit that makes the rest of the work trustworthy.

STATISTICS

 ± 3.71 dB understates the spread by an order of magnitude

You report across-fold standard deviation and justify it on the grounds that pooling rows "would understate variance given within-fold correlation". It runs the other way here: averaging ~29 rows per fold before taking the standard deviation shrinks it. Heart SAR is 6.46 ± 3.71 in your table; pooled across rows it is 6.30 ± 39.38 . A 39 dB spread on $n=145$ means the mean is not a usable summary — report medians, or bootstrap CIs once S2-04 lands, and always state which level the dispersion is computed at.

LITERATURE

Five "[Author(s) needed]" placeholders are all sitting in the Reading List

The report says full citation details "were not available at the time of writing". They are in the Notion Reading List, with DOIs:

- `aidriven` — Torabi, PhD thesis, McMaster, 2025
- `spectrotemporal` — Yaqub et al., *CMES*, 2025, 10.32604/cmcs.2025.071571
- `nmfcrnn` — Han, Quan, Matuszewski & Corbett, *Sensors*, 2026, 10.3390/s26134268
- `edgelung` — Puneet, Shankar, Koluguri & Srivastava, *IEEE BioCAS*, 2025
- `ssa` — Han & Quan, *IEEE ICSP*, 2025, 10.1109/ICSPS66615.2025.11347745

Thirty minutes of work, and it changes how the report reads.

LITERATURE

You read Yaqub and missed why it is on the list

Your related-work table lists `spectrotemporal` as "PhysioNet 2016, no separation, binary normal/abnormal classification". That is Yaqub et al. — and their experiment 4, where accuracy collapses from **89% on clean heart sounds to 41% on computationally separated ones**, is the single result this entire project exists to explain. The Reading List entry says so in as many words. Your report treats it as an unrelated classification paper.

Two related gaps follow from the same cause. You mark the Han & Quan dataset as "not stated — needs confirming"; the charter tells you it is 50 synthetic mixtures built from 10 cardiac and 5 respiratory recordings with 2% RMS Gaussian noise, and that distinction is the basis of C1's novelty claim. And your table gives `nmfcrnn`'s dataset as ICBHI + Fraiwan, where the charter has Han et al. learning dictionaries from HLS-CMDS isolated recordings — one of those is wrong and it is worth resolving, because that paper is the source of the S3-02 hyperparameters.

SCOPE

PROTOCOL.md drifts from the chartered research question

Your memo asks whether separating before classifying reduces accuracy relative to classifying isolated ground truth. The charter's C2 asks something sharper: sweep separation quality continuously and find the *knee point* — the SDR below which separation actively hurts. Your version reproduces Yaqub's finding under better controls; the charter's version explains it. Keep the charter's framing. The knee point is what makes the figure worth publishing.

HOW YOU WORKED

PROCESS

Work from the board and the papers, not from summaries

Your `BACKLOG.md` says: "The task that asked for the eval harness referenced 'the same triplet-level split as S2-03' — no ticket/spec with that ID exists anywhere in this repo." S2-03 is a row in the Notion Sprint Backlog, with a description, a deliverable and a due date. Its neighbours S3-02 and S3-04 carry the full reproduction hyperparameters and the leakage trap you rediscovered independently.

Every issue in the section above traces to the same root: the Yaqub miss, the placeholder citations, the unverified Han & Quan setup, the scope drift, the dataset pulled from GitHub instead of Mendeley. You built from task text and paper summaries without opening the charter or the Reading List. The work you produced from that position is genuinely impressive, which is the point — with the actual source material in front of you it would have been impressive *and* pointed at the right target.

Concretely, before the next work session: open the charter page, the Sprint Backlog and the Reading List, and keep them open. The "Why It Matters" column exists to tell you what to look for in each paper before you read it.

PROCESS

Keep the board current

Notion still shows S1-02 through S1-06 and all of S2 as "Not started", and you have finished most of them. I plan sprints from that board. Move your tickets as you go, and if you finish ahead — which you clearly will — tell me rather than starting the next thing, so I can point you at what is actually most useful next.

NEXT ACTIONS

1 Re-download HLS-CMDS from Mendeley

(`data.mendeley.com/datasets/8972jxbmp/3`) into a directory outside the git repo. Record version, download date, URL and per-file SHA-256 in `README.md`. This closes S0-06 and S1-01 properly.

- 2 **Re-run the additivity test on the Mendeley copy** and report back to me with one number: how many of the 145 rows satisfy `mixed = a · (heart+lung)` . Check the sample rate at the same time. This is the input to my C1 decision and it blocks kickoff planning, so do it first.
- 3 **Promote the additivity check into the repo** as a function in `load_dataset.py` — something like `verify_additive_triplets()` returning the valid row set — with a test. Your `verify_mix_triplets()` currently checks that files exist and IDs match; it should also check that the signals are actually related.
- 4 **Re-issue the results table** on whatever the valid subset turns out to be, with medians alongside means, and revise the section 7 interpretation. The +5.18 dB against Han & Quan's 5.7 dB is worth stating explicitly as a harness sanity check.
- 5 **Fill in the five citations** from the Reading List and re-read Yaqub with the 89→41% result in mind.
- 6 **Fix the repo:** register or remove the embedded `HLS-CMDS` git repository, and commit `baselines.py` — it is still untracked.

To be clear about the balance here: you delivered four sprints of infrastructure in the onboarding week, the split is better than what the charter specified, and you found the one anomaly in the data that matters. The corrections above are about aiming, not ability. Bring the Mendeley numbers to kickoff on Thursday and we will re-plan C1 around what is actually there.

Findings verified by re-running `metrics.py` , `split.py` and `baselines.py` against the dataset on server 230, plus independent additivity, cross-correlation, exhaustive-pairing and clipping-mechanism checks. Reported numbers reproduced exactly (heart SDR −13.73 vs −13.68 published).